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TemporalExpertNet: Cross-Temporal Knowledge Reuse for Promotion-Aware CVR Prediction

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Abstract

Major promotional events such as Black Friday and 618 Shopping Day cause drastic, heterogeneous shifts in user and advertiser behavior, posing persistent challenges for conversion rate (CVR) models trained on daily data. Existing methods often lack the flexibility to capture this periodic variability, resulting in poor modeling of diverse behavioral patterns. To address these challenges, we propose **TemporalExpertNet(TEN)**, a cross-temporal transfer learning framework for industrial-scale CVR prediction during promotion cycles. TEN decomposes the model into a stable representation encoder and a promotion-sensitive expert, enabling reusable temporal knowledge transfer. Specifically, we propose BridgeNet to address the mismatch between historical knowledge and current features through temporal representation alignment. We further introduce TemporalExpertGate (TEG) to perform sample-aware expert fusion, enabling fine-grained prediction adjustment and adaptive knowledge reuse across promotion periods. By using a two-stage training strategy, TEN achieves stable alignment and adaptive expert fusion for robust prediction under shifting promotional distributions. TEN was deployed on a large-scale short-video ads platform during the 618 preheating phase, improving conversion rate by 7.52% and platform RPM by 4.27% with only 0.23% model size and 1.8% latency overhead. It was therefore fully launched to all traffic on 618 Shopping Day, bringing substantial commercial gains.

CCS Concepts

• Information systems → Recommender systems.

*This work was done during an internship at Kuaishou Technology.

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1 Introduction

Major promotional events such as Black Friday¹ and the 618 Shopping Day² have become core revenue drivers for e-commerce platforms. While these events span less than 10% of the calendar year, they account for over 30% of the annual revenue for large-scale advertising platforms, highlighting their substantial commercial importance. However, this commercial revenue comes with a significant technical challenge: these events introduce sharp, heterogeneous shifts in user and advertiser behavior over short time spans, posing a critical challenge for conversion rate (CVR) prediction.

As illustrated in Figure 1, promotional periods exhibit heterogeneous dynamics on both the advertiser and user sides. Some advertisers substantially increase budgets and adjust targeting configurations, causing pronounced distributional shifts in ad delivery. Meanwhile, user behavior varies widely: some users show heightened conversion intent driven by price sensitivity, whereas others maintain stable daily patterns. These heterogeneous dynamics violate the i.i.d. assumptions underpinning most predictive models, resulting in considerable prediction bias and miscalibration when daily-trained CVR models are deployed during promotions. Consequently, platforms may suffer inefficient ad spend and degraded user experience precisely when reliability is most critical for revenue.

¹[https://en.wikipedia.org/wiki/Black_Friday_\(shopping\)](https://en.wikipedia.org/wiki/Black_Friday_(shopping))

²https://en.wikipedia.org/wiki/618_shopping_day

Despite its importance, this issue has been addressed by only a limited number of studies. Most existing CVR models, such as ESM [27] and MMoE [26], do not explicitly tackle the challenges posed by large-scale promotional scenarios. A recent method [3] attempts to alleviate distributional drift by fine-tuning on historical promotion data. However, they tend to overfit to outdated distributions and fail to differentiate between shifting and stable user behaviors, ultimately degrading performance for user segments less responsive to promotion dynamics. These limitations highlight a core challenge in developing a unified framework that can adaptively handle both behavioral drift and heterogeneity.

We observe that although behavioral drift is a universal phenomenon, promotional events themselves exhibit strong *temporal regularity*, such as their annual recurrence. This observation leads to our central research question: Can we develop a unified CVR prediction framework that adaptively distinguishes between stable and shifting behavior patterns, while supporting effective knowledge reuse across temporally disjoint yet semantically aligned distributions? To address this, we draw inspiration from modular transfer paradigms, which have demonstrated strong generalization across diverse tasks [7, 19]. These paradigms commonly fine-tune task-specific heads atop reusable encoders to enhance adaptability. We further validate the feasibility of this modular design for our setting through a structural analysis in Section 4.2, confirming its suitability for capturing stable promotional knowledge in our scenario.

Following this principle, we propose a structurally modular framework tailored for cross-temporal CVR prediction under non-stationary user behavior. We instantiate this as **TemporalExpertNet (TEN)**, which decouples the prediction model into reusable and event-specific components, and adaptively fuses their outputs through a novel representation alignment and expert routing mechanism. Specifically, TEN's core consists of three synergistic modules: (1) a structural decoupling mechanism that separates a stable promotion expert from a dynamic daily encoder; (2) **BridgeNet**, a behavior-aware alignment module that maps daily embeddings into the semantic space of the promotion expert, addressing the representational gap; and (3) **TemporalExpertGate (TEG)**, a sample-wise routing mechanism that adaptively fuses expert outputs based on contextual signals, tackling behavioral heterogeneity at the decision level. These modules are jointly optimized through a **two-stage training strategy**, designed to stabilize cross-temporal alignment and expert fusion under shifting behavioral distributions. By combining structural decoupling with behavior-aware transfer, TEN provides a principled and industrially viable solution for CVR prediction under non-stationary traffic environments.

The main contributions of this work are summarized as follows:

- We propose TEN, a novel cross-temporal CVR prediction framework that explicitly models dual behavior regimes through structural decoupling and enables effective knowledge transfer across non-stationary distributions.
- We design BridgeNet for representation alignment and TemporalExpertGate for sample-aware expert fusion, coupled with a two-stage training strategy to ensure stable alignment and expert fusion under shifting promotion distributions.
- We deploy TEN during the preheating phase of the 618 Shopping Day (2025) on a large-scale short-video ads platform, achieving

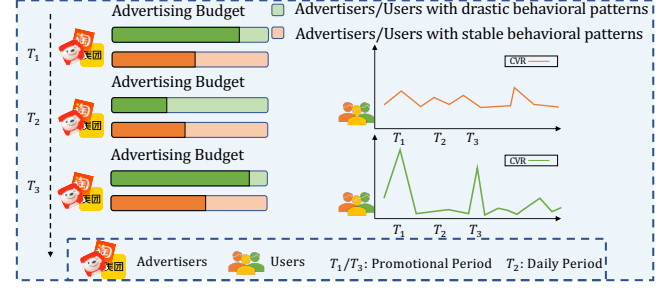


Figure 1: Cross-Temporal Behavioral Divergence

conversion lift of +7.52%, and RPM improvement of +4.27% over production baselines with only a +0.23% increase in model size and +1.8% inference latency overhead. Subsequently, TEN was fully launched to the entire platform traffic on 618 Shopping Day, delivering substantial business gains.

2 Related Work

2.1 Conversion Rate Prediction

CVR prediction is a core task in industrial recommender and advertising systems, as it directly influences revenue optimization and user engagement [11, 12, 26, 27, 33, 42]. Early models largely reused CTR prediction architectures, combining linear components with deep networks to capture low- and high-order feature interactions [6, 12]. To better model user interests, subsequent work introduced sequential architectures with attention over behavior histories [32, 41, 42], and later multi-interest or Transformer-based encoders to capture diverse and temporally sensitive user intents [2, 20, 37]. A parallel line of work studies delayed feedback modeling, where conversions arrive long after clicks. Early methods explicitly model the delay distribution [4], while later online approaches treat unobserved samples as negatives and relabel them upon conversion, typically with importance sampling to correct bias [18, 35]. More recent studies argue that duplicating positives distorts the data distribution and instead duplicate real negatives with refined reweighting [5, 9]. These approaches generally assume a relatively stable delay distribution, an assumption that may hold during regular periods but often breaks down during major promotions, where both user and advertiser behaviors become highly non-stationary. In contrast, our framework explicitly decouples daily and promotion-specific components and is designed to remain robust under such temporally volatile regimes.

2.2 Transfer Learning

In recommender systems, distribution shift can be naturally viewed as a domain shift between different feature distributions, where user behaviors and item representations vary across environments. Transfer learning [8, 10, 15, 17, 21, 30, 36, 44] addresses this challenge by leveraging auxiliary knowledge from a source domain to improve generalization in a target domain. Early instance-based and feature-based methods [31, 46] reduce domain mismatch by reweighting source samples or aligning latent user and item spaces. Parameter-based strategies, such as consensus regularization and

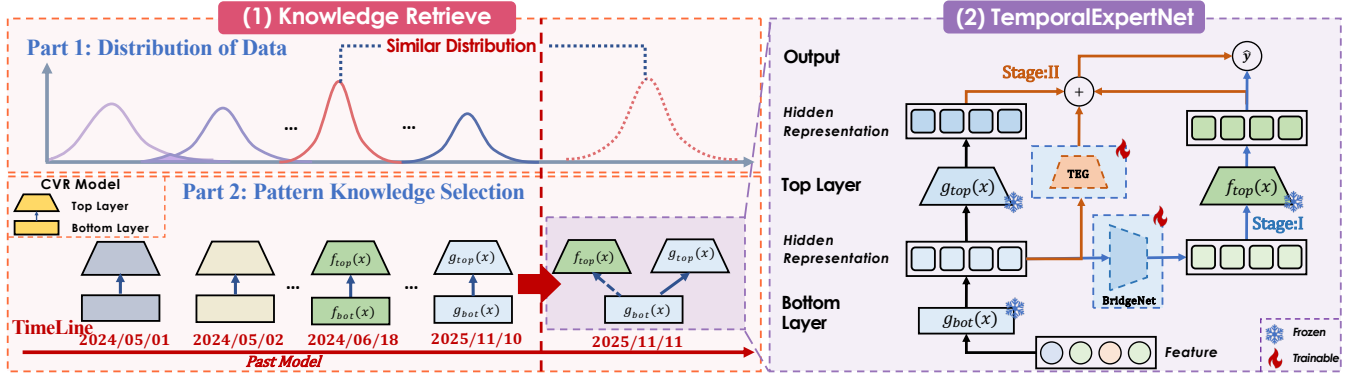


Figure 2: Overview of TemporalExpertNet (TEN). The framework integrates daily behavior encoders and historical promotion experts via BridgeNet for cross-temporal representation alignment and TemporalExpertGate for sample-aware expert fusion. A two-stage training strategy further stabilizes the adaptation and routing process

Bayesian neural transfer [13, 45], further enable modular reuse of pretrained components across tasks. Cross-domain recommendation frameworks [14, 22–24, 28, 29, 34, 38–40, 43] advance this line of work by aligning user-item embeddings and interaction semantics across distribution-shifted domains. However, most existing approaches focus on static domain gaps [16, 25] and fail to capture the dynamic behavioral shifts induced by recurring promotion events, where the domain boundary itself evolves over time and static transfer mechanisms often break down. HDR [3] introduces a bias adaptation module trained on historical promotion data to address distributional drift, but it overlooks behavioral heterogeneity. These limitations motivate our development of *TemporalExpertNet*, a structured transfer learning framework that explicitly decouples stable representations from behavior-variant response modules and performs expert selection via context-aware routing.

3 Methodology

We propose TemporalExpertNet (TEN), a cross-temporal transfer framework designed to address behavioral non-stationarity in large-scale promotional scenarios. The framework decouples the predictive model into a shared-bottom encoder and multiple scenario-specific expert modules, enabling the flexible reuse of historical promotional knowledge. This knowledge, which includes high-level behavioral patterns learned from past campaigns, is transferable to similar future events. As illustrated in Figure 2, the TEN architecture facilitates effective cross-temporal knowledge transfer by dynamically recombining a daily encoder with a reusable promotion expert. To mitigate the semantic misalignment introduced by this recombination, we introduce **BridgeNet**, a module designed to align the representation space of the daily encoder with the semantic space expected by the promotion expert. To address behavioral heterogeneity, the **TemporalExpertGate (TEG)** module performs sample-aware expert fusion, adaptively routing each input based on contextual signals. To ensure robust performance and avoid gradient interference, we employ a **two-stage training strategy** that sequentially optimizes these components, stabilizing both representation alignment and expert routing.

3.1 Architectural Principles

To robustly adapt to dynamic traffic regimes, it is essential to first analyze how user responses evolve across temporal scenarios. Major promotional events induce significant shifts in conversion behavior, which compromise the generalization capabilities of models trained on daily data.

We formalize this challenge by starting from the standard factorization of a joint probability distribution:

$$p(y, \mathbf{x}) = p(y | \mathbf{x}) \cdot p(\mathbf{x}) \quad (1)$$

In real-world advertising platforms, promotional periods often directly follow daily periods, leaving little time for substantial changes in key covariates. User attributes such as geographic location, demographics, and long-term interests therefore remain largely stable, suggesting an approximately stable marginal feature distribution:

$$p_{\text{daily}}(\mathbf{x}) \approx p_{\text{promo}}(\mathbf{x}) \quad (2)$$

However, the conditional response distribution, which captures the decision-making process given the same features, often shifts markedly during promotions:

$$p_{\text{daily}}(y | \mathbf{x}) \neq p_{\text{promo}}(y | \mathbf{x}) \quad (3)$$

This conditional shift reflects the fact that users may become more impulsive, price-sensitive, or responsive to discounts during promotions, leading to significantly different conversion behaviors. Such shifts violate the i.i.d. assumption underlying most predictive models, and account for the bias and miscalibration commonly observed when daily-trained CVR models are directly deployed in promotion scenarios.

Despite this volatility, promotions across different campaigns often exhibit a degree of **structural regularity**. Our empirical analysis (Section 4.2) also proves that conditional patterns learned from one campaign can generalize to another, as their underlying promotion-specific semantics remain highly consistent.

$$p_{\text{promo}}^{\text{old}}(y | \mathbf{x}) \approx p_{\text{promo}}^{\text{new}}(y | \mathbf{x}) \quad (4)$$

Based on this insight, we propose a **knowledge reuse** strategy. Instead of training a fresh model for every campaign, we can transfer high-level promotional knowledge from a previously learned model, provided the transfer mechanism accounts for the current feature distribution and behavioral heterogeneity.

3.1.1 Design Principle: Decoupling Reusable Knowledge from Volatile Behavior. To achieve this, we separate representation learning from scenario-specific response modeling. For a historical promotion model, we can express the prediction function as:

$$M_{\text{promo}}(\mathbf{x}) = f_{\text{top}}(f_{\text{bot}}(\mathbf{x})), \quad \mathbf{x} \sim \mathcal{D}_{\text{promo}}, \quad (5)$$

where f_{bot} is the feature encoder and f_{top} is the promotion-specific expert. Similarly, for the current daily model, we define:

$$M_{\text{daily}}(\mathbf{x}) = g_{\text{top}}(g_{\text{bot}}(\mathbf{x})), \quad \mathbf{x} \sim \mathcal{D}_{\text{daily}}, \quad (6)$$

where g_{bot} is the daily encoder and g_{top} is the daily expert.

3.1.2 Cross-Temporal Hybridization Path. This leads to our **cross-temporal hybridization path**. During a new campaign, we form a hybrid transfer path by injecting the up-to-date daily representation into the historical promotion expert:

$$M^*(\mathbf{x}) = f_{\text{top}}(g_{\text{bot}}(\mathbf{x})), \quad \mathbf{x} \sim \mathcal{D}_{\text{promo}}^{\text{new}} \quad (7)$$

This approach realizes the knowledge reuse objective without re-training the entire stack. The daily encoder g_{bot} remains fresh and adapted to current traffic, while the promotion expert f_{top} contributes stable, promotion-aware semantics accumulated from prior campaigns.

3.1.3 Challenges and Architectural Solutions. This hybridization path, while promising, introduces two significant challenges that must be addressed for the framework to be effective:

- **Representation Mismatch:** The latent space generated by $g_{\text{bot}}(\mathbf{x})$ under daily training differs from that expected by the historical f_{top} , since $\mathcal{D}_{\text{daily}} \neq \mathcal{D}_{\text{promo}}$. Formally,

$$g_{\text{bot}}(\mathbf{x}) \sim f_{\text{bot}}(\mathbf{x}) \quad (8)$$

This mismatch can degrade the predictive performance of f_{top} when it receives misaligned inputs.

- **Heterogeneous Responsiveness:** Not all samples exhibit promotional behavior patterns. Some users maintain their daily patterns even during campaigns, while others shift their behavior sharply. A single expert is thus insufficient. Our architecture must therefore provide a mechanism to adaptively handle these different user behaviors.

In summary, the presence of conditional shift and cross-campaign regularity motivates a structurally decoupled architecture. To realize this, our framework requires an alignment interface to reconcile latent spaces and a sample-aware routing mechanism to account for behavioral heterogeneity.

3.2 BridgeNet: Behavior-Aware Feature Alignment Across Temporal Domains

To enable effective knowledge reuse under temporal distribution shifts, we propose BridgeNet, a behavior-aware alignment module that reconciles semantic differences between daily and promotional representations. It combines global projection of daily embeddings into the promotion expert's space with interaction-level fusion via a

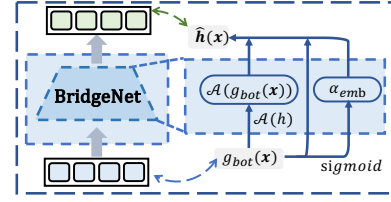


Figure 3: Detailed architecture of BridgeNet

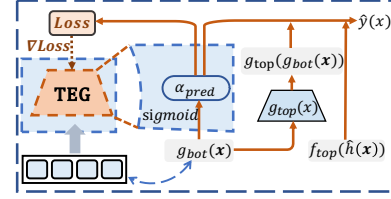


Figure 4: Detailed architecture of TemporalExpertGate

gating mechanism. This design addresses both global representation misalignment and fine-grained drift, supporting accurate and robust transfer across temporal domains, as illustrated in Figure 3.

3.2.1 Global Semantic Alignment. To reconcile the representational discrepancy between the frozen daily encoder g_{bot} and the frozen historical promotion head f_{top} , BridgeNet introduces a lightweight transformation layer $\mathcal{A}(\cdot)$. This layer performs preliminary alignment by projecting the frozen daily embeddings into the semantic space expected by the promotion expert:

$$M^*(\mathbf{x}) = f_{\text{top}}(\mathcal{A}(g_{\text{bot}}(\mathbf{x}))) \quad (9)$$

$$\mathcal{A}(\mathbf{h}) = \mathbf{W}_2 \cdot \sigma(\mathbf{W}_1 \cdot \mathbf{h} + \mathbf{b}_1) + \mathbf{b}_2 \quad (10)$$

Unlike conventional domain adaptation pipelines that rely on auxiliary supervision, this projection layer is jointly trained with the CVR objective. Since f_{top} and g_{bot} remain frozen, the transformation is implicitly guided to align the daily encoder outputs with the semantic expectations of the promotion expert.

3.2.2 Interaction-Level Fusion. To selectively adapt promotion knowledge when necessary, we introduce a soft gating mechanism that blends the original and adapted embeddings based on interaction-specific drift. The final fused representation is defined as:

$$\tilde{\mathbf{h}}(\mathbf{x}) = \alpha_{\text{emb}} \cdot \mathcal{A}(g_{\text{bot}}(\mathbf{x})) + (1 - \alpha_{\text{emb}}) \cdot g_{\text{bot}}(\mathbf{x}) \quad (11)$$

The gating function $\alpha_{\text{emb}} \in [0, 1]$ is computed as:

$$\alpha_{\text{emb}} = \text{sigmoid}(\mathbf{W}_g \cdot g_{\text{bot}}(\mathbf{x}) + \mathbf{b}_g) \quad (12)$$

Therefore, the final output of the promotion expert becomes:

$$M^*(\mathbf{x}) = f_{\text{top}}(\tilde{\mathbf{h}}(\mathbf{x})) \quad (13)$$

This enables the model to adaptively control the transfer strength per interaction, activating the promotion pathway only when the user-ad context exhibits sufficient deviation from daily behavior.

3.3 TemporalExpertGate: Sample-Aware Expert Fusion for Output-Level Adaptation

While BridgeNet facilitates representation-level alignment across temporal domains, it does not address the challenge of determining the appropriate contribution of each expert, whether daily or promotional, toward the final prediction. To address this, we introduce TemporalExpertGate (TEG), a sample-aware routing module that performs output-level adaptation by dynamically blending expert predictions based on contextual signals, as illustrated in Figure 4.

3.3.1 Expert Routing via Gated Fusion. User responses during promotion periods are highly heterogeneous. Some users demonstrate abrupt behavioral shifts driven by urgency or discounts, while others retain patterns similar to their daily routines. Using a single expert across all samples fails to account for behavioral heterogeneity, often leading to poorly calibrated predictions. TEG mitigates this issue by computing a soft interpolation of expert predictions. Given the fused representation $\tilde{\mathbf{h}}(\mathbf{x})$ from BridgeNet, the final prediction is:

$$\hat{y}(\mathbf{x}) = \alpha_{\text{pred}} \cdot f_{\text{top}}(\tilde{\mathbf{h}}(\mathbf{x})) + (1 - \alpha_{\text{pred}}) \cdot g_{\text{bot}}(g_{\text{bot}}(\mathbf{x})) \quad (14)$$

where f_{top} and g_{top} are the promotion and daily experts, respectively, and $\alpha_{\text{pred}} \in [0, 1]$ is a learnable gating weight defined as:

$$\alpha_{\text{pred}} = \text{sigmoid}(\mathbf{W}_o \cdot (g_{\text{bot}}(\mathbf{x})) + \mathbf{b}_o) \quad (15)$$

This mechanism enables TEG to dynamically adjust expert contributions based on sample-specific behavioral context, supporting fine-grained adaptation to temporal drift.

3.3.2 Confidence-Regularized Expert Specialization. To enhance expert disentanglement and avoid degenerate routing where the gating network produces near-uniform outputs, we introduce a confidence regularization term that biases the gate towards decisive expert selection. Concretely, we define:

$$\mathcal{L}_{\text{reg}} = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{\text{promo}}^{\text{new}}} [\alpha_{\text{pred}} (1 - \alpha_{\text{pred}})], \quad (16)$$

where $\alpha_{\text{pred}} \in [0, 1]$ denotes the gating probability assigned to the promotion expert. This quadratic term attains its maximum at $\alpha_{\text{pred}} = 0.5$ and decreases monotonically as α_{pred} approaches either extreme (0 or 1). By minimizing $\mathcal{L}_{\text{reg}}$ jointly with the CVR loss, the gating network is encouraged to produce high-confidence assignments, thereby routing each instance predominantly to a single expert rather than averaging their predictions.

Such decisive routing imposes a strong inductive bias that encourages clear separation between daily and promotion experts, allowing each to specialize in its respective behavioral regime without interference. This specialization mitigates the dilution of expert-specific patterns in heterogeneous traffic and ensures that samples strongly aligned with a given regime are routed to the most relevant expert, thereby enhancing adaptation accuracy.

3.4 Two-Stage Training Strategy

To ensure stable and effective cross-temporal knowledge reuse, we propose a two-stage training strategy that sequentially optimizes the alignment and routing components. Direct joint optimization is unstable due to the severe representation mismatch between the current daily encoder and the historical promotion expert. In early training, the daily expert is much easier to optimize, which causes

Table 1: Data statistics across different stages.

Scenario	#Users	#Items	#Samples
Daily (avg.)	~40M	~500K	~70M
Preheating (avg.)	~60M	~600K	~90M
Promotion Day (avg.)	~80M	~700K	~110M

the gating network to quickly bias toward the daily branch, while the promotion expert receives weak and noisy gradients. This leads to ineffective semantic alignment and unstable expert routing. The two-stage design is to explicitly decouple representation alignment from expert fusion, thereby preventing such gradient interference before the promotion expert becomes semantically compatible with the current daily encoder.

3.4.1 Stage I Representation Alignment with Frozen Promotion Expert. In the first stage, all components except BridgeNet are frozen, including g_{bot} , g_{top} , f_{top} and TEG. BridgeNet is trained to align daily representations to the semantic space of the promotion expert, with the loss computed solely from the output of f_{top} :

$$\hat{y}^{(1)}(\mathbf{x}) = f_{\text{top}}(\tilde{\mathbf{h}}(\mathbf{x})), \quad \mathcal{L}^{(1)} = \mathcal{L}_{\text{CE}}(\hat{y}^{(1)}(\mathbf{x}), y) \quad (17)$$

This step prevents the promotion expert from being directly exposed to out-of-distribution daily representations and establishes semantic compatibility between the aligned representation $\tilde{\mathbf{h}}(\mathbf{x})$ and the historical promotion expert, which is critical for stabilizing subsequent expert routing.

3.4.2 Stage II Expert Fusion and Routing Optimization. Once the representation is aligned, we fix BridgeNet and train the TemporalExpertGate to learn expert fusion. At this stage, both experts operate on semantically meaningful representations, which enables the gating network to learn routing decisions based on predictive reliability rather than optimization shortcuts. The final prediction is a weighted combination of the two experts, where the weights sum to one and are adaptively determined:

$$\hat{y}^{(2)}(\mathbf{x}) = \alpha_{\text{pred}} \cdot f_{\text{top}}(\tilde{\mathbf{h}}(\mathbf{x})) + (1 - \alpha_{\text{pred}}) \cdot g_{\text{top}}(g_{\text{bot}}(\mathbf{x})) \quad (18)$$

The total loss includes the CVR prediction loss and a regularization term to prevent collapse of the gating mechanism:

$$\mathcal{L}^{(2)} = \mathcal{L}_{\text{CE}}(\hat{y}^{(2)}(\mathbf{x}), y) + \lambda \cdot \mathcal{L}_{\text{reg}} \quad (19)$$

This two-stage strategy ensures that expert routing is learned on top of semantically aligned inputs, effectively mitigating gradient interference and improving prediction robustness under distributional shifts in promotional traffic.

4 Experiment

4.1 Experimental Setup

4.1.1 Dataset. Due to the lack of publicly available datasets capturing the fine-grained traffic dynamics of large-scale promotional events, we construct an industrial-scale dataset from impression logs of a commercial short-video advertising platform. All models are trained on regular daily traffic data to ensure stability in standard serving environments. As shown in Table 1, we then evaluate effectiveness through a three-stage protocol covering peak promotion days, preheating periods, and full-scale deployment.

- **Offline Evaluation (11.11):** We first evaluate our framework on data collected from November 11, 2024, the peak day of the Double 11 Shopping Festival. As shown in Table 1, this dataset contains over 100 million samples in a single day and reflects extreme user behavior shifts and traffic volatility. This evaluation setting simulates the worst-case domain shift scenario and offers a rigorous offline test of our model’s generalization ability.
- **Preheating Phase Validation:** To assess the model’s adaptability during the gradual traffic transition preceding a major event, we further use real-time data from May 15–20, 2025, corresponding to the preheating phase of the 618 Shopping Day. Compared to regular daily traffic, this phase exhibits increased user activity and behavior volatility, providing a middle-ground testbed that reflects partially shifted but still mixed behavior patterns.
- **Full-Scale Online Deployment (6.18):** Our model was fully deployed on the platform during the peak day of the 618 Shopping Day(2025), which serves as the final stage of evaluation. The online deployment covers traffic from hundreds of millions of Daily Active Users (DAUs), demonstrating the framework’s readiness for real-world use at scale.

4.1.2 Evaluation Metrics. To comprehensively assess the performance of TEN, we adopt a multi-faceted evaluation protocol, encompassing ranking quality, calibration accuracy, and real-world business effectiveness.

- **AUC[1].** We frame CVR prediction as a binary classification task and report the Area Under the ROC Curve (AUC)³ to quantify the model’s discriminative power between conversion and non-conversion events.
- **PCOC.** To assess prediction calibration, we adopt Predicted Conversion over Conversion (PCOC), defined as the ratio of predicted CVR to actual CVR, i.e., $PCOC = \frac{\text{predicted CVR}}{\text{actual CVR}}$. A value closer to 1 indicates better calibration.
- **LOSS & RMSE.** To further assess the numerical fidelity of predicted probabilities, we report Cross-Entropy Loss (LOSS) and Root Mean Squared Error (RMSE). These metrics reflect how closely the predicted values approximate ground truth labels.
- **CVR & RPM.** In online A/B testing, we evaluate business impact through two key metrics. Conversion Rate (CVR) reflects how well the system matches users’ conversion intent and directly contributes to advertiser-side benefits. Revenue Per Mille (RPM) corresponds to platform-side revenue, defined as the expected revenue generated per one thousand ad impressions.

4.1.3 Implementation Details. A key challenge in deploying TEN lies in selecting an appropriate historical promotion expert head as the source of transferable knowledge. The pool of candidate models is diverse, encompassing various campaign nodes (e.g., 618 Shopping Day, Double 11 cycles). Following the approach in [3], we first construct day-level statistical features for each historical sample batch. The analysis reveals that typical large-scale promotions exhibit high similarity in user behavior distributions, with elevated structural feature similarity scores, indicating a strong statistical basis for knowledge transfer. Based on this observation, we fix the model transfer from June 18, 2024 as the offline knowledge injection source and use data from June 17–18, 2024 for fine-tuning the

transfer process. We set the regularization weight λ to 1 in practice to ensure balanced optimization during transfer fine-tuning.

4.1.4 Compared Methods. To comprehensively evaluate the effectiveness and practicality of TEN, we compare it against a set of strong baselines and ablated variants. Our selection of baselines is guided by real-world industrial relevance and aims to validate our approach against current best practices for handling both temporal drift and feedback delay. In particular, we include the most advanced existing method [3], along with promotion-specific heuristics and state-of-the-art delayed feedback modeling (DFM) algorithms. These baselines span both industrial production setups and academic solutions, ensuring that the evaluation reflects realistic deployment constraints while also benchmarking against cutting-edge research.

- **Base.** The standard *online production baseline* in our advertising system, trained exclusively on recent daily traffic via incremental updates. It follows the MMoE architecture [26] and works well in stable environments.
- **Base w/promotion transfer.** A direct cross-temporal transfer baseline that applies the model trained on a previous promotion (e.g., 618) directly to the current event (e.g., Double 11) without further adaptation. This approach preserves promotion-specific behavioral patterns learned from the source event.
- **Base w/promotion indicator.** A variant of the Base Model augmented with manually selected promotion-related features (e.g., page styles). This baseline introduces promotion context at the feature level without modifying the model architecture.
- **HDR.** A state-of-the-art fine-tuning framework tailored for promotion scenarios [3]. HDR leverages historical promotion-period data to adapt daily models by selectively updating promotion-relevant submodules.
- **ES-DFM.** A delay feedback modeling algorithm[35] that optimizes the expected distribution of real transformations through importance sampling. It estimates the importance weight of each training sample and uses the estimated weights to correct the loss function in model training.
- **FNW.** A delay feedback modeling algorithm[18] that initializes all samples as negative and reuses positive samples with corrected labels once feedback arrives, incorporating importance weights into the cross-entropy loss to adjust for the bias between the observed and true distributions.
- **TEN w/o Two-Stage.** A variant that jointly trains BridgeNet and TEG from scratch instead of using the proposed two-stage optimization strategy.
- **TEN w/o TEG.** A variant that retains the representation alignment module (BridgeNet) but removes the sample-wise expert routing mechanism (TEG), applying the promotion expert uniformly to all samples.
- **TEN w/o BridgeNet.** A variant that removes the BridgeNet and directly connects the current daily encoder to the historical promotion expert. The expert fusion module (TEG) is retained, but the semantic spaces are not aligned.

4.2 Empirical Motivation

To validate the feasibility of our modular design and the cross-temporal transferability of behavioral knowledge, we conduct an

³In online advertising systems, an AUC increase of 0.1% is a significant improvement.

Table 2: Layer-wise Parameter Comparison Across Time

Short-Term Gap: 11/11 to 11/28					Long-Term Gap: 11/11 to 12/12				
Level	Date	Mean	Std	Sim	Level	Date	Mean	Std	Sim
Top	11/11	0.0013	0.1890	0.9935	Top	11/11	0.0013	0.1890	0.9908
	11/28	0.0012	0.1896		Top	12/12	0.0012	0.1906	
Bot	11/11	-0.0074	0.2167	0.9685	Bot	11/11	-0.0074	0.2167	0.9444
	11/28	-0.0074	0.2210		Bot	12/12	-0.0074	0.2264	

empirical analysis of model parameters and user behavior across two temporal scenarios: (i) a short-term shift from a major promotional event (November 11) to a regular day (November 28), and (ii) a long-term shift between two semantically aligned promotions (November 11 and December 12, a smaller-scale promotional event). Our investigation unfolds across three key dimensions: (1) parameter similarity and statistical characteristics across different model layers, (2) cross-temporal generalization performance of models trained on different dates, and (3) consistency of user behavior across distinct promotional campaigns.

4.2.1 Representation Stability Across Temporal Gaps. Our model decouples into a bottom layer for low-level feature encoding (e.g., embeddings and feature interactions) and a top layer for high-level response prediction (e.g., final MLP and output head). As shown in Table 2, we report the cosine similarity and parameter-level statistics (mean and standard deviation) for both the top and bottom layers. We find that the bottom layers exhibit pronounced temporal drift over longer intervals. Their cosine similarity drops from 0.9685 in the short-term comparison to 0.9444 in the long-term case, and standard deviation increases from 0.2210 to 0.2264. These results indicate that bottom-layer representations are highly sensitive to recency effects and undergo notable shifts even across similarly structured promotion events. In contrast, the top layers remain remarkably stable. Cosine similarity only slightly decreases from 0.9935 to 0.9908, and the standard deviation remains nearly constant (0.1890 to 0.1906). This suggests that top-layer parameters encode high-level promotional intent patterns that generalize well across temporally disjoint but behaviorally aligned campaigns.

4.2.2 Cross-Temporal Behavioral Consistency. A naive expectation is that a more recently trained model should generalize better to a nearby test set. To probe the actual drivers of transferability, we compare three models on the December 12 test day, with the results shown in Table 3a: a model from a past promotion (November 11), a temporally closer daily model (November 28), and a same-day model (December 12). While the same-day model performs best, confirming the importance of a fresh bottom encoder, a notable counter-intuitive finding emerges: the November 11 promotion model significantly surpasses the November 28 daily model in both ranking and calibration, despite being chronologically more distant. This is particularly striking given that the bottom-layer parameters of the November 28 model are in fact more similar to those of the December 12 test set. This suggests that the promotion-specific top layer from the November 11 model captures high-level behavioral patterns that remain relevant for December 12, a factor more critical than low-level feature recency. To further substantiate this, we compute the Pearson correlation coefficients of key aggregate statistics (e.g., conversion rate, impression volume) between training days.

As shown in Table 3b, the correlation between November 11 and December 12 is 0.46, substantially higher than the 0.20 correlation between November 28 and December 12. This reinforces that behavioral alignment across campaigns is driven by latent structural similarities in promotion dynamics, rather than mere chronological proximity. This collective evidence validates TEN’s design of combining a fresh bottom encoder with a reusable, promotion-aware top expert.

Table 3: Correlation and Performance Metrics on 12/12

(a) Model Performance			(b) Data Similarity			
Model	AUC	PCOC		11/11	11/28	12/12
11/11	0.9092	0.8445	11/11	1.00	-0.39	0.46
11/28	0.8870	0.8320	11/28	-0.39	1.00	0.20
12/12	0.9444	0.9976	12/12	0.46	0.20	1.00

4.3 Overall Performance

Table 4 compares all methods in Section 4.1.4 across ranking, calibration, and numerical fidelity metrics. The **Base Model** reaches an AUC of 0.9403 on promotional data but suffers from poor calibration, with a PCOC of only 0.9193, indicating systematic underestimation of conversion probability. Adding historical promotion knowledge directly in **Base w/promotion transfer** essentially applies a straightforward transfer learning strategy, where a model trained on past promotion data is directly reused in a new campaign. However, in large-scale advertising systems this naive approach often fails due to severe representational drift between source and target promotions, as reflected here by the lowest AUC of 0.9367 and an overestimated PCOC of 1.0664. Incorporating a binary promotion flag in **Base w/promotion indicator** slightly improves AUC to 0.9409 and PCOC to 0.9210, but static feature augmentation alone remains insufficient to capture the rapidly evolving behavioral shifts of promotion traffic.

We next evaluate delayed feedback modeling approaches, which are widely used in advertising systems to correct for label bias from conversion delays—a bias that is further amplified during promotions. Both **ES-DFM** and **FNW** achieve higher overall performance than the Base model, but their AUC scores of 0.9415 and 0.9412 remain far below TEN’s 0.9456. The performance gap stems from their reliance on stable delay distributions to estimate importance weights, an assumption often broken during promotions when user decisions become more impulsive and volatile. The method explicitly tailored for promotion adaptation, **HDR**, employs a global feature alignment strategy across time, achieving an AUC of 0.9435 and a PCOC of 1.0575. This represents a clear improvement over the previous baselines.

In contrast, **TEN** delivers the best overall results, achieving a superior AUC of 0.9456, a well-controlled PCOC of 1.0562, and the lowest RMSE and LOSS. These results validate the effectiveness of TEN’s structural decoupling and expert reuse design. By separating stable and volatile behavioral components in the representation space and enabling sample-aware expert routing, TEN leverages historical promotion knowledge while adapting to real-time user

intent. This fine-grained transfer capability enables accurate and calibrated predictions under non-stationary traffic regimes, translating into significant commercial gains.

Table 4: Comparison of Model Variants

Model Variant	AUC	RMSE	LOSS	PCOC
Base	0.9403	0.1768	0.1161	0.9193
Base w/promotion transfer	0.9367	0.1869	0.1271	1.0664
Base w/promotion indicator	0.9409	0.1769	0.1159	0.9210
HDR	0.9435	0.1768	0.1155	1.0575
ES-DFM	0.9415	0.1768	0.1159	0.9324
FNW	0.9412	0.1769	0.1160	0.9487
TemporalExpertNet (Full)	0.9456	0.1765	0.1144	1.0562
w/o Two-Stage	0.9441	0.1766	0.1145	1.0639
w/o TEG	0.9440	0.1766	0.1148	1.0626
w/o BridgeNet	0.9425	0.1772	0.1164	0.9363

4.4 Ablation Study

As shown in Table 4, to investigate the contribution of each component in our proposed framework, we conduct a systematic ablation study by removing key modules within TemporalExpertNet.

Removing the two-stage training pipeline (w/o *Two-Stage*) decreases AUC to 0.9441 and raises LOSS and PCOC, indicating that joint optimization of BridgeNet and TEG from scratch introduces gradient interference, undermining cross-temporal alignment. Excluding the expert routing mechanism (w/o *TEG*) further weakens performance, reducing AUC to 0.9440 and increasing PCOC to 1.0626. This validates the importance of sample-aware expert fusion in managing heterogeneous user behaviors during promotions. Finally, when BridgeNet is disabled (w/o *BridgeNet*), the model suffers from the largest performance drop among all variants, with AUC reduced to 0.9425. This demonstrates that direct reuse of the promotion head without proper representation alignment fails to handle semantic mismatch between the daily encoder and historical expert. Collectively, these results highlight that each component of our design is indispensable for achieving robust performance under non-stationary promotion environments.

4.5 Online Deployment

We conducted a staged online deployment of TEN on a large-scale short-video advertising platform to evaluate its robustness under non-stationary traffic. The deployment began during the preheating phase of the 618 Shopping Day(2025) in mid-May, allowing for initial validation of its performance as promotion-driven user activity began to rise. Following consistent positive results, TEN was fully deployed on June 18, subjecting it to the year’s highest-traffic and most volatile conditions.

4.5.1 Deployment Pipeline. The deployment procedure, illustrated in Figure 5, follows a structured initialization and fine-tuning strategy. TEN is first warm-started from a daily model trained on recent non-promotional traffic, activating the daily encoder and head. The promotion expert is initialized using top-layer parameters from a

historical model trained on logs from November 10–11, 2024 (Double 11 period), while BridgeNet and TEG are randomly initialized. To enable effective knowledge reuse, we perform two-stage fine-tuning on the imported promotion expert. In Stage I, we freeze TEG and optimize BridgeNet using the same historical logs, aligning the daily encoder to the reused promotion head. In Stage II, we fix BridgeNet and activate TEG to learn sample-wise expert fusion based on contextual drift.

4.5.2 Online Results and Observations. We conduct an online A/B test with 25% real-time traffic allocated to the experimental group. Due to traffic constraints, the control group adopts the second-best offline baseline HDR, while the experimental group deploys TEN. All other components of the ad-serving pipeline remain unchanged. Over the 6-day window, TEN consistently outperforms the baseline. The weighted average CVR uplift reaches +7.52%, and the weighted average RPM improvement reaches +4.27%. Notably, the largest gains occur on May 15 and 16, coinciding with the abrupt launch of the 618 preheating phase. During this period, user–ad interaction patterns deviate sharply from daily traffic. By leveraging historical promotional knowledge and dynamically routing samples under behavioral drift, TEN yields immediate gains on May 15–16 and remains stable on May 18–19 as traffic returns closer to routine patterns. This demonstrates TEN’s robustness and strong generalization across both volatile and stable periods, underscoring the effectiveness of cross-temporal knowledge reuse. By leveraging expert routing, TEN achieves precise adaptation across heterogeneous user behaviors, addressing the limitations of prior models under mixed-distribution settings.

4.5.3 Full Deployment on 618 Shopping Day. Following the promising results in the online A/B testing phase, TEN was fully deployed to 100% of real-time traffic on a platform with over **hundreds of millions of Daily Active Users (DAUs)** during the 618 Shopping Day(2025). This large-scale deployment placed the model in direct contact with massive and highly dynamic user–advertiser interactions, where any performance gain could translate into substantial commercial impact. The 618 Shopping Day is characterized by extreme surges in both user engagement and advertiser competition, resulting in a highly volatile and challenging environment for conversion modeling. Traffic volume peaks multiple times within a single day, with rapid shifts in user intent and ad inventory composition. Figure 7 illustrates the relative lifts in CVR and RPM over the course of the day, aggregated in four-hour intervals. In Figure 7a, TEN delivers consistent conversion improvements across all time windows, peaking at **+8.0%** during the 08:00–12:00 slot while maintaining elevated gains even during lower-traffic intervals, reflecting

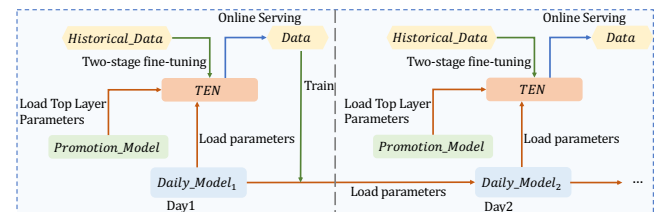


Figure 5: System Deployment and Training Pipeline

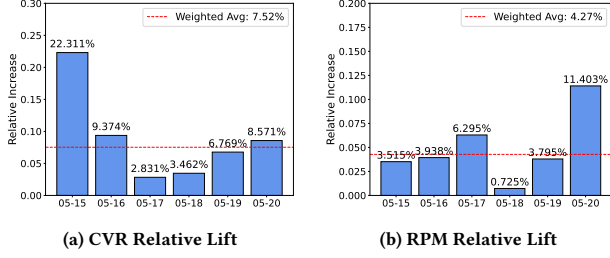


Figure 6: Comparison of relative lift in CVR and RPM

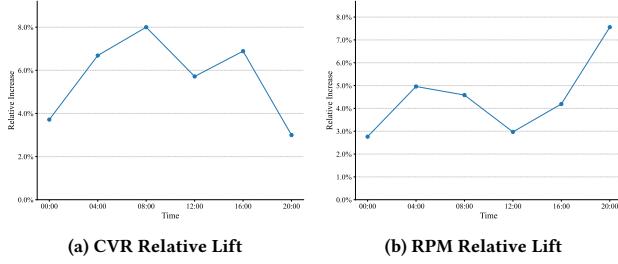


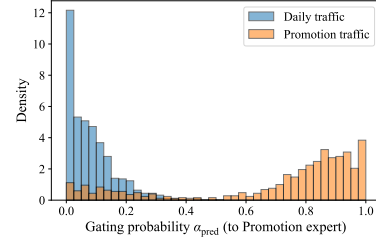
Figure 7: Relative lift in CVR and RPM during 618 Shopping Day under full deployment of TEN

the model’s ability to retain promotional knowledge and respond to transient shifts in user intent. In Figure 7b, RPM shows a similar trend, reaching up to +7.6% improvement in the evening peak and sustaining a positive lift throughout the day, demonstrating that the uplift in CVR effectively translates into tangible monetization gains, benefiting both advertisers and the platform. Both metrics remain above baseline across all segments of the day, even during off-peak intervals, demonstrating TEN’s ability to sustain high-quality predictions and commercial gains during one of the most competitive e-commerce events of the year.

4.5.4 Model Complexity and Inference Latency. Despite introducing additional modules for cross-temporal transfer, TEN is designed for minimal computational and latency overhead. In industrial-scale advertising models, the vast majority of parameters are concentrated in the bottom layers, primarily in feature embeddings, due to extremely high feature cardinality. TEN’s modifications are strategically confined to the upper layers, resulting in only a marginal +0.23% increase in total model size. In terms of inference latency, TEN incurs just a +1.8% increase over the base model, remaining well within production tolerances. Consequently, TEN delivers substantial performance gains while fully satisfying the strict latency and throughput requirements of large-scale ad serving systems.

4.6 Qualitative Analysis of Expert Routing

To qualitatively examine TEG’s routing behavior, we sample instances from a regular day (Nov 28) and a promotional day (Nov 11) and visualize the distribution of gating probabilities α_{pred} , which indicate the model’s confidence in selecting the promotion expert

Figure 8: Gating probabilities α_{pred} for different traffic

(values near 1) versus the daily expert (values near 0). As shown in Figure 8, the distributions in the two settings are clearly polarized, demonstrating that confidence regularization effectively enforces sharp expert separation across traffic regimes. On regular days, most samples concentrate near $\alpha_{\text{pred}} \approx 0$, whereas during promotions, the peak shifts toward $\alpha_{\text{pred}} \approx 1$. A small portion of promotional samples are still routed to the daily expert, reflecting stable user behavior even under promotion conditions. Overall, TEG achieves effective expert specialization and sample-level adaptive routing, capturing both regime-level differences and within-regime behavioral heterogeneity.

5 Conclusion

This paper proposes **TemporalExpertNet (TEN)**, a modular and promotion-adaptive framework for robust CVR prediction under non-stationary user behavior during major commercial campaigns. TEN decouples the prediction architecture into stable daily encoders and promotion-sensitive experts, enabling reusable knowledge transfer across temporally disjoint traffic regimes. Through the integration of **BridgeNet** for cross-temporal representation alignment and **TemporalExpertGate (TEG)** for sample-aware expert fusion, the framework achieves fine-grained adaptation, while a **two-stage training strategy** stabilizes the alignment and routing process across temporal boundaries. Deployed on a large-scale short-video ads platform, TEN first demonstrated its real-world utility during the preheating phase of the 618 Shopping Day (2025), achieving +7.52% CVR and +4.27% RPM improvements over production baselines. Building on this success, TEN was **fully launched to all traffic** on the peak day, delivering substantial commercial gains and demonstrating its ability to maintain high performance under the most demanding, large-scale deployment conditions. Looking ahead, this work opens avenues for future research, including generalizing expert modularity to multi-event scenarios, incorporating causal routing signals to mitigate confounded behavior drift, and extending temporal fusion strategies to multi-task or retention-based modeling. We believe TEN not only provides a deployable solution for industrial CVR prediction but also lays the groundwork for structured transfer learning in temporally evolving recommendation systems.

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Ethical Considerations

Our work on **TemporalExpertNet (TEN)** aims to advance the state-of-the-art in CVR prediction for commercial platforms, with the goal of improving ad relevance for users and efficiency for advertisers. However, we recognize that as with any machine learning system operating on user data, it is crucial to consider the potential ethical implications, including issues of fairness, privacy, and broader societal impact.

Fairness and Bias. Predictive models trained on user interaction logs risk inheriting and amplifying societal biases, potentially creating feedback loops where certain demographic groups are shown fewer or lower-quality ads. In deploying TEN, we actively monitor and mitigate such risks through a dedicated fairness evaluation pipeline. This includes segment-wise AUC and calibration checks across gender, age, and regional cohorts, along with drift detection on feature distributions to identify disproportionate shifts. Any statistically significant performance gap between segments triggers re-ranking adjustments in the downstream delivery pipeline, where exposure weights are rebalanced to ensure equitable ad distribution without degrading overall system performance.

User Privacy. All input features are anonymized at the source by hashing user and advertiser identifiers. Sensitive attributes are excluded from training, and data retention follows strict internal policies that comply with regional data protection laws. Feature engineering operates on aggregated or time-windowed statistics to prevent user re-identification. Model checkpoints are version-controlled and audited to ensure no introduction of personally identifiable information.

Societal Impact. The effectiveness of a CVR model inevitably raises concerns about potential over-influence on user decision-making. A highly optimized system could, in theory, nudge users toward purchases they might not have made otherwise. In designing and deploying TEN, our objective is to enhance user experience by aligning ad delivery more closely with genuine user intent, while improving the overall efficiency of the ad ecosystem for advertisers. To uphold this commitment, we enforce practical safeguards in deployment, including frequency capping, diversity constraints in ad exposure, and real-time anomaly monitoring to detect and mitigate overly aggressive targeting. These measures ensure that TEN operates within the boundaries of responsible AI principles, serving the best interests of both users and the broader society.

Mitigation and Future Work. This research was conducted with these ethical considerations embedded into both model design and deployment. TEN already incorporates fairness monitoring, privacy-compliant data handling, and adjustments to mitigate bias in ad delivery. Looking ahead, we plan to extend these safeguards by integrating fairness-aware learning objectives directly into training, adopting stronger privacy-enhancing technologies and developing transparent auditing tools to continuously assess societal impact.

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